

FILIP RUZICKA

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EMPLOYMENT

- 2024– Marie Skłodowska-Curie fellow, Institute of Science and Technology Austria (Vienna, Austria), with Prof. Beatriz Vicoso
- 2019–23 Postdoc (Monash University, Australia), with A/Prof. Tim Connallon

EDUCATION

- 2015–18 PhD, Evolutionary Genetics (University College London, UK), with Profs. Max Reuter and Kevin Fowler
- 2012–13 MRes Biodiversity, Evolution & Conservation (Institute of Zoology & Natural History Museum, London, UK)
- 2009–12 Bsc Biological Sciences (University College London, UK)

PUBLICATIONS

- 2025 **Ruzicka, F.***, Zwoinska, M.K.*, Goedert, D., Kokko, H., Li Richter X-Y., Moodie, I.R., Nilén, S., Olito, C., Svensson, E.I., Czappon, P., Connallon, T. A century of theories of balancing selection. *Biological Reviews* (*in press*)
- Connallon, T., Czappon, P., Olito, C., Goedert, D., Kokko, H., Nava-Bolaños, A., Nilén, S., Svensson, E.I., Zwoinska, M.K., Dutoit, L.*, **Ruzicka, F.***. Predicting the prevalence of genetic trade-offs among adaptive substitutions. *Evolution* 24:qpaf061
- 2024 McDonough, Y., **Ruzicka, F.**, Connallon, T. Reconciling theories of dominance with the relative rates of adaptive substitution on sex chromosomes and autosomes. *Proceedings of the National Academy of Sciences U.S.A.* 121(44): e2406335121
- 2022 **Ruzicka, F.**, Holman, L., Connallon, T. Polygenic signals of sex differences in selection in humans from the UK Biobank. *PLOS Biology*, 20:e3001768
- Ruzicka, F.**, Reuter, M. Evolutionary Genetics: Dissecting a sexually antagonistic polymorphism. *Current Biology (Dispatch)*, 35:PR828-R830
- Connallon, T., Beasley, I.J., McDonough, Y., **Ruzicka, F.** How much does the unguarded X contribute to sex differences in lifespan? *Evolution Letters* 6:319–329
- Ruzicka, F.**, Connallon, T. An unbiased test reveals no enrichment of sexually antagonistic polymorphisms on the human X chromosome. *Proceedings of the Royal Society B: Biological Sciences* 289:202123142
- 2021 **Ruzicka, F.**, Connallon T., Reuter, M. Sex differences in deleterious mutational effects in *Drosophila melanogaster*: combining quantitative and population genetic insights. *Genetics* 219:iyab143
- Jardine, M., **Ruzicka, F.**, Diffley, C., Fowler, K., Reuter, M. A non-coding indel polymorphism in the *fruitless* gene of *Drosophila melanogaster* exhibits antagonistically pleiotropic effects. *Proceedings of the Royal Society B: Biological Sciences* 288:20202958
- 2020 **Ruzicka, F.***, Connallon, T.*. Is the X chromosome a hot spot for sexually antagonistic polymorphisms? Biases in current empirical tests of classical theory. *Proceedings of the Royal Society B: Biological Sciences* 287:20201869

	Ruzicka, F.* , Dutoit, L.* , Czuppon, P., Jordan, C.Y., Li, X-Y, Olito, C., Runemark, A., Svensson, E.I., Yazdi, H.P., Connallon, T. The search for sexually antagonistic genes: Practical insights from studies of local adaptation and statistical genomics. <i>Evolution Letters</i> 4:398-415
2019	Ruzicka, F.* , Hill, M.S.* , Pennell, T.M.* , Flis, I., Ingleby, F.C., Mott, R., Fowler, K., Morrow, E.H., Reuter, M. Genome-wide sexually antagonistic variants reveal long-standing constraints on sexual dimorphism in fruit flies. <i>PLOS Biology</i> 17:e3000244
2018 and previous	Connallon, T., Olito, C., Dutoit, L., Papoli, H., Ruzicka, F. , Yong, L. Local adaptation and the evolution of inversions on sex chromosomes and autosomes. <i>Philosophical Transactions of the Royal Society B: Biological Sciences</i> 373:20170423
	Reuter, M., Camus, M. F., Hill, M.S., Ruzicka, F. , Fowler, K. (2017). Evolving plastic responses to external and genetic environments. <i>Trends in Genetics (Forum)</i> 33:169-170
	Davies, T., Ruzicka, F. , Lavery, T., Walters, C.L., Pettoirelli, N. (2016). Ultrasonic monitoring to assess the impacts of forest conversion on Solomon Island bats. <i>Remote Sensing in Ecology and Conservation</i> 2:107-118
	Andujar, C., Arribas, P., Ruzicka, F. , Crampton-Platt, A., Timmermans, M.J.T.N., Vogler, A.P. (2015). Phylogenetic community ecology of soil biodiversity using mitochondrial metagenomics. <i>Molecular Ecology</i> 24:3603-3617
Manuscripts (available on request)	Ruzicka, F. , Cyza, J., Connallon, T. A genome-wide negative correlation between the severity and dominance of human deleterious mutations.
	Mackintosh, C.* , Ruzicka, F.* , Connallon, T. Trade-offs and the evolution of dominance during selective sweeps.
	Ruzicka, F. , Vicoso, B. Testing a new theory for faster-X adaptation in <i>Drosophila melanogaster</i> .

*denotes shared authorship

FUNDING & AWARDS

2023	€134,260, IST BRIDGE Fellowship (Horizon Europe, Marie-Skłodowska Curie Actions, #101034413)
2020	A\$1,000, ECR Award (Australasian Evolution Society)
2019	A\$5,000, ECR “Evolution Theme” Leader (Monash School of Biological Sciences)
	A\$10,000, ECR Career Development Award (Monash School of Biological Sciences)
2018	Runner-up, ECR Talk, the UK Genetics Society’s Autumn Symposium “Genotype to Phenotype to Fitness”
	Winner, Final Year Talk, UCL GEE Postgraduate Symposium
2017	£800, attendance 26 th ESEB Congress (UCL Graduate School Fund)
	Winner, Second Year Poster, UCL GEE Postgraduate Symposium
2016	£3,000, CEE Summer Symposium (London Centre for Ecology & Evolution, the UK Genetics Society)
2015	CHF450, “Evolutionary Biology in Guarda” Summer School (University of Basel)
	£7,500, Further PhD research funds (UK Natural Environment Research Council)
2014	£64,000, PhD Studentship (UK Natural Environment Research Council)
2013	“Darwin Prize” for Top Student, UCL MRes programme

LEADERSHIP

2024–	Co-organiser, “EvoLunch” Seminar Series (ISTA)
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2022–23	Co-organiser, “Genetics and Evolution of Sex Differences” symposium (Society for Molecular Biology & Evolution)
2020–21	Co-organiser, School of Biological Sciences Seminar Series (Monash)
2019–21	ECR representative, School of Biological Sciences “Evolution” Theme (Monash)
2016–18	Co-organiser, GEE Friday Seminars for Young Researchers (UCL)
2016	Co-organiser, London Centre for Ecology & Evolution Summer Symposium: “War & Peace: The Dynamics of Evolutionary Conflict” (London)

TEACHING

2024–25	PhD tutor, “Population genetics: the basics” PhD course (ISTA)
2020–21	Co-founder, “Foundations of Evolutionary Biology” PhD course in population and quantitative genetics (Monash)
2019–21	Guest lecturer, Evolutionary & Ecological Genetics (Monash), Genomics & Population Genetics (Monash)
2015–21	Undergraduate tutor, Genomics & Population Genetics (Monash), Evolutionary & Ecological Genetics (Monash), Sex Genes & Evolution (UCL), Computational Biology (UCL), Evolutionary Genetics (UCL)

SUPERVISION

2021–22	Naarah Bovingdon (Honours, Monash) Yasmine McDonough (Honours, Monash) Natalie Schmiedl (Undergraduate, Monash)
2020	Justyna Cyza (Undergraduate, Monash)
2019	Isobel Beasley (Undergraduate, Monash)
2018	Olivia Davidson (MRes, UCL)
2016	Harvi Pawar (Undergraduate, UCL) Didem Snaith (Undergraduate, UCL) Max Houghton (MRes, UCL)
2015	Sadé Bates (Msci, UCL)

PRESENTATIONS

2025	Talk, EvolVienna (Vienna, Austria) Talk, 58 th Population Genetics Group (Sheffield, UK)
2024	Invited talk, International Laboratory for Human Genome Research, National Autonomous University of Mexico (online) Talk, 3 rd Joint Congress for Evolutionary Biology (Montréal, Canada) Talk, UCL GEE Joint General Meeting (London, UK) Poster, Probabilistic Modelling in Genomics (Vienna, Austria)
2023	Invited talk, Edinburgh University (Edinburgh, UK) Invited talk, Lund University (Lund, Sweden) Invited talk, Institute of Science and Technology Austria (Vienna, Austria) Poster, Jacques Monod Conference “Sex unfolded” (Roscoff, France) Talk, 56 th Population Genetics Group (London, UK)
2021	Talk, Australasian Evolution Society (Melbourne, Australia)
2020	Plenary talk, Australasian Evolution Society (online) Talk, Monash Genetics Genomics & Health seminar series (Melbourne, Australia)
2019	Talk, Australasian Evolution Society (Sydney, Australia) Talk, Monash ECR Science Symposium (Melbourne, Australia) Talk, Monash School of Biological Sciences seminar series (Melbourne, Australia) Poster, 27 th European Society for Evolutionary Biology (Turku, Finland) Talk, Genetics Society of Australasia (Melbourne, Australia) Talk, 52 nd Population Genetics Group (Oxford, UK)
2018	Talk, The Genetics Society’s Autumn Symposium “Genotype to Phenotype to Fitness” (Exeter, UK)

- Talk, UCL GEE Postgraduate Symposium (London, UK)
 Talk, 51st Population Genetics Group (Bristol, UK)
 2017 Talk, 26th European Society for Evolutionary Biology (Groningen, Netherlands)
 Poster, UCL GEE Postgraduate Symposium (London, UK)
 2015 Talk, European Meeting of PhD Students in Evolutionary Biology (Stirling, UK)
 Poster, Quantitative Genomics (London, UK)
 Talk, UCL GEE Postgraduate Symposium (London, UK)
 Talk, NERC DTP Panel (UCL)

REVIEWING

- 2019– *PNAS, Nature Ecology and Evolution, PLOS Biology, PLOS Genetics, Molecular Biology and Evolution, The American Naturalist, Proceedings B, Genome Biology and Evolution, Evolution Letters, Genetics, Evolution, Journal of Evolutionary Biology, Behavioural Ecology and Sociobiology, Heredity*
 National Science Foundation (USA), Human Frontiers Science Program
 Postdoctoral Fellowships, The Austrian Academy of Sciences DOC fellowships (Austria)

PUBLIC ENGAGEMENT

- 2020 Writer, *Evolution Letters* research blog
 2018 Scientific expert, 'Agonism/Antagonism' art exhibition at UCL Grant Museum of Zoology
 2016–17 Writer, *The Genetics Society* Summer newsletter
 2014–17 Tutor, "Generating Genius" volunteer programme aimed at STEM secondary school students from low SES backgrounds
 2009–13 Leader, "Roots & Shoots" volunteer programme to communicate scientific and environmental issues to primary school children

TRAINING & WORKSHOPS

- 2017–24 ESEB Special Topics Network (Lund, Sweden; Montpellier, France; Ingelheim am Rhein, Germany)
 2019–20 GENETICS Peer Review Training Programme (online)
 2015 Guarda Summer School (Guarda, Switzerland)

REFEREES

- Beatriz Vicoso Professor, Institute of Science and Technology Austria (Vienna, Austria). **E:** bvicoso@ista.ac.at
 Tim Connallon A/Professor, School of Biological Sciences, Monash University (Melbourne, Australia). **E:** Tim.connallon@monash.edu
 Max Reuter Professor, Dept. of Genetics Evolution & Environment, University College London (London, UK). **E:** m.reuter@ucl.ac.uk